CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 4: **Carbon and its Compounds** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Carbon compounds have low melting and boiling points. Reason (R): Intermolecular forces in covalent compounds are weak. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. The mass of CO_2 formed on complete combustion of 16 g methane (C = 12, H = 1, O = 16) is: 1

(a) 44 g **(b)** 88 g **(c)** 22 g **(d)** 16 g

3. A hydrocarbon X decolourises bromine water and has molar mass 28 g/mol. X is: 1

(a) ethyne **(b)** methane **(c)** ethene **(d)** ethane

4. Assertion (A): Unsaturated hydrocarbons burn with a sooty flame. Reason (R): They have a higher percentage of carbon so burn incompletely. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

5. The mass of water formed on burning 16 g methane is: 1

(a) 18 g **(b)** 36 g **(c)** 72 g **(d)** 16 g

6. An organic compound Y gives effervescence with $NaHCO_3$ and is used as a preservative. Y is: 1

(a) ethanoic acid **(b)** methanal **(c)** ethanol **(d)** ethane

7. Assertion (A): Diamond is a poor conductor of electricity but graphite conducts. Reason (R): In graphite, each carbon is bonded to four others. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

8. The volume of CO_2 at STP on burning 1 mol ethane completely is: 1

(a) 89.6 L **(b)** 67.2 L **(c)** 22.4 L **(d)** 44.8 L

9. Compound Z on heating with conc. H_2SO_4 gives ethene and on oxidation gives ethanoic acid. Z is: 1

(a) methanol **(b)** ethane **(c)** ethanol **(d)** ethanal

10. Assertion (A): Soap does not form lather in hard water. Reason (R): Hard water contains calcium and magnesium ions which form insoluble salts with soap. 1

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Calculate the mass of CO_2 formed on burning 30 g of ethane (C = 12, H = 1, O = 16). 2

12. An alkane has 22 covalent bonds. Find its molecular formula. 2

13. How many moles of H_2 are needed to saturate 1 mol of propyne? 2

14. Calculate the percentage of carbon in ethane. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A hydrocarbon contains 85.7% carbon and 14.3% hydrogen, and its molar mass is 42. Find the formula and name it. 3

16. Determine the number of C-C and C-H bonds in butane and in isobutane and comment. 3

17. State the effect of chain length on physical state in the alkane series with examples. 3

OR

Calculate the volume of oxygen at STP needed to burn 2 mol of methane and the volume of CO_2 produced. 3

18. Explain why graphite conducts but diamond does not, and why fullerene is a molecular solid. 3

OR

Compute the molecular mass of the 5th member of the alkane series and the difference between members. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain conversion of ethanol to ethene, ethanoic acid and ethyl ethanoate with reagents and equations. Identify the oxidising agent and the catalyst. 5

OR

Draw structures and name all isomers of C_4H_{10}, C_4H_8 positions (but-1-ene, but-2-ene) and explain the difference between structural isomers and homologues. 5

20. Calculate: 4.6 g of ethanol burns completely (C = 12, H = 1, O = 16). Find the mass of CO_2 and water, and volume of CO_2 at STP. 5

OR

Explain the allotropy of carbon with bonding diagrams (described), property and use of diamond, graphite, fullerene and why they burn to give the same product. 5

END OF QUESTION PAPER — Total: 40 Marks